Math211
Discrete Mathematics

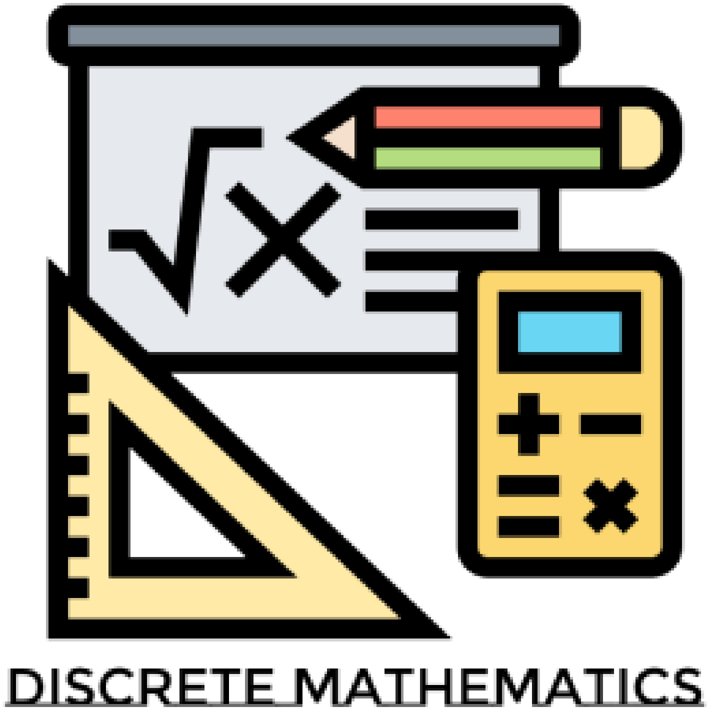
Propositional Logic II

Agenda

- 1.2 Applications of Propositional Logic
  - Translating English to logical expressions
- 1.3 Propositional Equivalences

Discrete Mathematics and Its Applications

Kenneth H. Rosen



1.2 Applications of Propositional Logic

Translating English to logical expressions

Why?

- English is often ambiguous and translating sentences into compound propositions removes the ambiguity
- Using **logical expressions**, we can analyze them and determine their truth values
- We can use rules of inferences to reason about them

Example 1

Let p = "It is hot", and q = "It is sunny"

It is **not** hot.	$\neg p$
It is hot **and** sunny.	$p \wedge q$
It is hot **or** sunny.	$p \vee q$
It is **not** hot **but** sunny.	$\neg p \wedge q$
It is **neither** hot **nor** sunny.	$\neg p \wedge \neg q$

Example 2

Conclusion

"You can access the internet from campus

if you are a computer science major or you are not a freshman."

Hypothesis

q : "You can access the internet from campus"

p : "You are a computer science major"

r : "You are freshmen"

$$(p \vee \neg r) \rightarrow q$$

H → C

System specification

- Translating sentences in natural language into **logical expressions** is an essential part of specifying both hardware and software systems
- Consistency of system specification

Example: System specification

"The automated reply **cannot** be sent **when** the file system is full"

1. Let q denote "The automated reply can be sent"
2. Let p denote "The file system is full"

The logical expression is

$$p \rightarrow \neg q$$

Converse, Inverse, Contrapositive

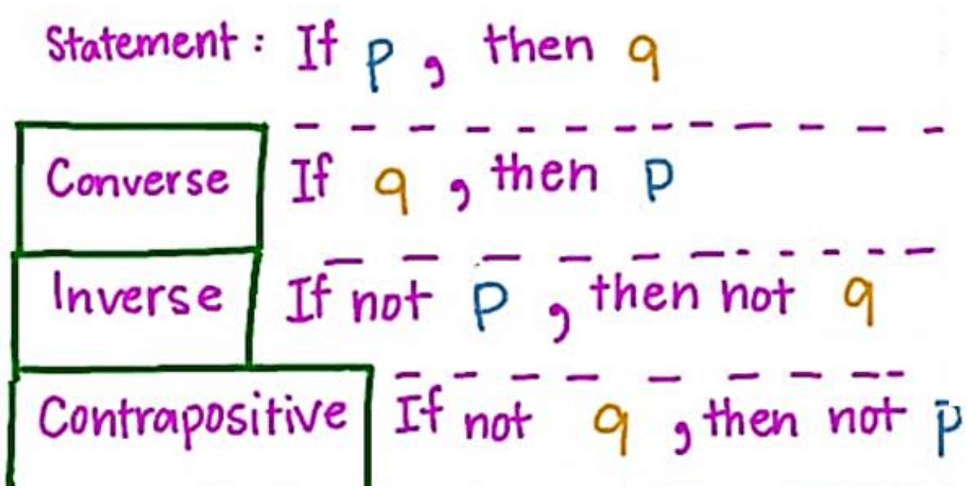
Some terminology, for an implication $p \to q$:

- Its *converse* is: $q \to p$
- Its *inverse* is: $\neg p \to \neg q$
- Its *contrapositive*: $\neg q \to \neg p$

switch the order

negate

switch & negate



p q

Read the following conditional statement: {If it is raining, then Amelia has her umbrella up.}
Write the converse of the statement. Write the inverse of the statement.
Write the contrapositive of the statement.

Statement: If p, then q

Converse	If q, then p	If Amelia has her umbrella up, then it is raining.
Inverse	If not p, then not q	If it is not raining, then Amelia does not have her umbrella up.
Contrapositive	If not q, then not p	If Amelia does not have her umbrella up, then it is not raining.

= hypothesis = p

= conclusion = q

One of these three has the *same meaning* (same truth table) as $p \to q$.

Can you figure out which?

Truth Table — Contrapositive

$p \to q$
statement

$q \to p$
converse

$\neg p \to \neg q$
inverse

$\neg q \to \neg p$
contrapositive

				Conditional	Converse	Inverse	Contrapositive
p	q	~p	~q	p→q	q→p	~p→~q	~q→~p
T	T	F	F	T	T	T	T
T	F	F	T	F	T	T	F
F	T	T	F	T	F	F	T
F	F	T	T	T	T	T	T

Example

$p \to q$	$q \to p$	$\neg p \to \neg q$	$\neg q \to \neg p$
statement	**converse**	**inverse**	**contrapositive**

If $x=2$, then $x^2=4$.

Statement ≡ Contrapositive

- The Statement If $x=2$, then $x^2=4$ is true.
- Its Contrapositive If $x^2 \neq 4$, then $x \neq 2$ so is true.

Converse ≡ inverse

- Its Converse If $x^2=4$, then $x=2$ is not true (x could be equal to -2)
- Its inverse If $x \neq 2$, then $x^2 \neq 4$ so is not true (x could be equal to -2)

Example

$p \to q$	$q \to p$	$\neg p \to \neg q$	$\neg q \to \neg p$
statement	**converse**	**inverse**	**contrapositive**

Statement ≡ Contrapositive
Converse ≡ inverse

"The stores are closed when it is a holiday"

If it is a holiday, then the stores are closed

p: It is a holiday   q: the stores are closed

Converse: If the stores are closed, then it is a holiday.

Inverse: If it is not a holiday, then the stores are not closed.

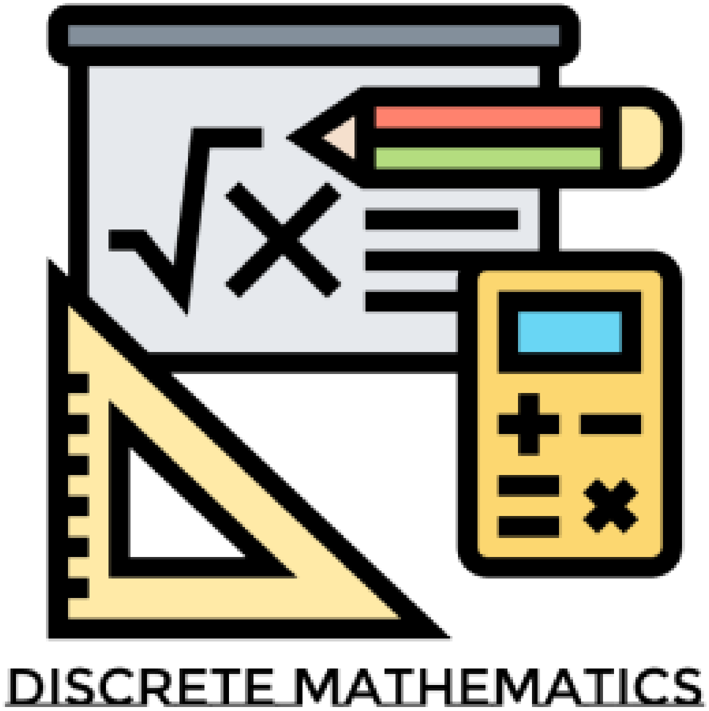
Contrapositive: If the stores are not closed, then it is not a holiday.

Example

$p \to q$	$q \to p$	$\neg p \to \neg q$	$\neg q \to \neg p$
statement	**converse**	**inverse**	**contrapositive**

If the sun is shining, then I go to store.

- Contrapositive: If I don't go to store then the sun is not shining.
- Converse: If I go to store then the sun is shining.
- Inverse: If the sun is not shining then I don't go to store



1.3 Propositional Equivalences

Propositional Equivalence

Two *syntactically* (*i.e.*, textually) different compound propositions may be the *semantically* identical (*i.e.*, have the same meaning). We call them *equivalent*.

Learn:

- Various *equivalence rules* or *laws*
- How to ***prove*** equivalences using *symbolic derivations*

Some Terminology

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T			
F			

Some Terminology

p	¬p	p ∨ ¬p	p ∧ ¬p
T	F	T	F
F	T	T	F

Contingency

Result is neither tautology nor contradiction

Tautology

Result is always true, no matter what p is

Contradiction

Result is always false, no matter what p is

Some Terminology

- A tautology is a proposition which is always true
  Classic Example: $p \vee \neg p$
- A contradiction is a proposition which is always false
  Classic Example: $p \wedge \neg p$
- A contingency is a proposition which neither a tautology nor a contradiction.
  Example: $(p \vee q) \rightarrow \neg r$

Logical Equivalence

- Two compound propositions r and s are *logically equivalent*

  if r $\leftrightarrow$ s is a tautology.

We write: r $\Leftrightarrow$ s or r $\equiv$ s

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

i.e. r and s have the same truth values as each other in all rows of their truth tables

Proving Equivalence via Truth Tables

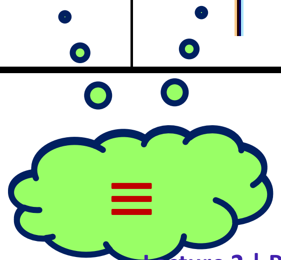
- Show that $A \leftrightarrow B$ and $\neg(A \oplus B)$ are equivalent

- Truth tables can be used to show the equivalence of two logical expressions

Proving Equivalence via Truth Tables

- Show that $A \leftrightarrow B$ and $\neg(A \oplus B)$ are equivalent

A	B	$A \oplus B$	$\neg(A \oplus B)$	$A \leftrightarrow B$
T	T	F	T	T
T	F	T	F	F
F	T	T	F	F
F	F	F	T	T



- Truth tables can be used to show the equivalence of two logical expressions
- The last two columns have the same values -- so the expressions are equivalent

Proving Equivalence via Truth Tables

$$p \to q \Leftrightarrow \neg p \vee q$$

Proving Equivalence via Truth Tables

$$p \rightarrow q \Leftrightarrow \neg p \vee q$$

Important for removing implication

p	**q**	**($p \rightarrow q$)**	**$\neg p$**	**$\neg p \vee q$**
F	F	T	T	T
F	T	T	T	T
T	F	F	F	F
T	T	T	F	T

Proving Equivalence via Truth Tables

Prove that $p \vee q \Leftrightarrow \neg(\neg p \wedge \neg q)$.

p	q	$p \vee q$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$	$\neg(\neg p \wedge \neg q)$
F	F					
F	T					
T	F					
T	T					

Proving Equivalence via Truth Tables

Prove that $p \vee q \Leftrightarrow \neg(\neg p \wedge \neg q)$.

p	q	$p \vee q$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$	$\neg(\neg p \wedge \neg q)$
F	F	F	T	T	T	F
F	T	T	T	F	F	T
T	F	T	F	T	F	T
T	T	T	F	F	F	T

Equivalence Laws

Equivalence	Name
$p \land T \equiv p$ $p \lor F \equiv p$	Identity
$p \land q \equiv q \land p$ $p \lor q \equiv q \lor p$	Commutative order
$p \lor (p \land q) \equiv p$ $p \land (p \lor q) \equiv p$	Absorption

Equivalence Laws

- *Domination:* $p \vee \mathbf{T} \Leftrightarrow \mathbf{T}$ $\quad p \wedge \mathbf{F} \Leftrightarrow \mathbf{F}$
- *Idempotent:* $p \vee p \Leftrightarrow p$ $\quad p \wedge p \Leftrightarrow p$
- *Double negation:* $\neg\neg p \Leftrightarrow p$
- *Associative:* $(p \vee q) \vee r \Leftrightarrow p \vee (q \vee r)$
  $(p \wedge q) \wedge r \Leftrightarrow p \wedge (q \wedge r)$

Grouping

More Equivalence Laws

- *Distributive:* $p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$
  $p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$
- *De Morgan's:*
  $$\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$$
  $$\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q$$
- *Trivial tautology/contradiction:*
  $$p \vee \neg p \Leftrightarrow \mathbf{T} \qquad p \wedge \neg p \Leftrightarrow \mathbf{F}$$

Defining Operators via Equivalences

- Exclusive Or

$$p \oplus q \Leftrightarrow (p \vee q) \wedge \neg(p \wedge q)$$
$$p \oplus q \Leftrightarrow (p \wedge \neg q) \vee (q \wedge \neg p)$$

- Implies

$$p \rightarrow q \Leftrightarrow \neg p \vee q$$

Equivalences with Conditionals and Biconditionals

Important for removing implication

- Conditionals
  - $p \rightarrow q \equiv \neg p \vee q$
  - $p \rightarrow q \equiv \neg q \rightarrow \neg p$
  - $\neg(p \rightarrow q) \equiv p \wedge \neg q$
- Biconditionals
  - $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
  - $p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$
  - $\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$

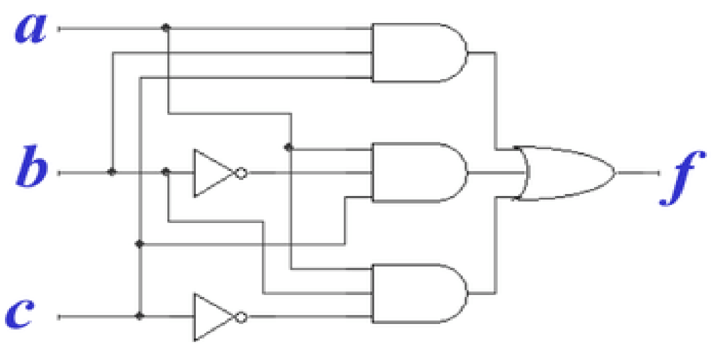
Defining Operators via Equivalences

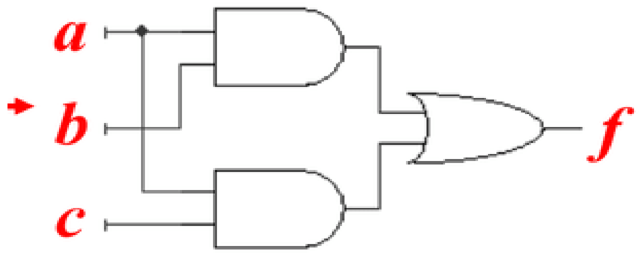
Using equivalences, we can *define* operators in terms of other operators.

$$(a \wedge b \wedge c) \vee (a \wedge \neg b \wedge c) \vee (a \wedge \neg b \wedge c)$$

$$\equiv$$

$$(a \wedge c) \vee (a \wedge b)$$





Manipulating Propositions

- **Compound propositions** can be simplified by using simple rules.
- Some are obvious, e.g. Identity, Domination, Idempotence, double negation, commutativity, associativity
- Less obvious: Distributive, De Morgan’s laws

Example 1

$p \leftrightarrow q \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$

$p \rightarrow q \Leftrightarrow \neg p \vee q$

Show that ¬(p → q) and p ∧¬q are logically equivalent using logic algebra.

$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws

$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

Example 1

Show that $\neg(p \to q)$ and $p \land \neg q$ are logically equivalent using logic algebra.

$$
\begin{aligned}
\neg(p \to q) &\equiv \neg(\neg p \lor q) && \text{by implies} \\
&\equiv \neg(\neg p) \land \neg q && \text{by the second De Morgan law} \\
&\equiv p \land \neg q && \text{by the double negation law}
\end{aligned}
$$

Example 2

$p \leftrightarrow q \Leftrightarrow (p \to q) \wedge (q \to p)$
$p \to q \Leftrightarrow \neg p \vee q$

Show that $p \to q \Leftrightarrow \neg q \to \neg p$ using logic algebra

$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws	$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws	$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$\neg(\neg p) \equiv p$	Double negation law	$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws	$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

Example 2

Show that $p \to q \Leftrightarrow \neg q \to \neg p$ using logic algebra

$p \to q$	
$\neg p \vee q$	removing implication
$\neg p \vee \neg(\neg q)$	$\neg\neg x = x$ (double negation of q)
$\neg(\neg q) \vee \neg p$	commutative law
$(\neg q) \to (\neg p)$	Implication $x \to y \Leftrightarrow \neg x \vee y$

Example 3

$p \leftrightarrow q \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$

$p \rightarrow q \Leftrightarrow \neg p \vee q$

- **Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology using logical algebra**

$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws

$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

Example 3

- **Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology using logical algebra**

$(p \wedge q) \rightarrow (p \vee q) \equiv \neg(p \wedge q) \vee (p \vee q)$	Implies Law
$\equiv (\neg p \vee \neg q) \vee (p \vee q)$	De Morgan law
$\equiv (\neg p \vee p) \vee (\neg q \vee q)$	Associative and commutative law for disjunction
$\equiv \mathrm{T} \vee \mathrm{T}$	Commutative law for disjunction
$\equiv \mathrm{T}$	Domination law

Example 4

$p \leftrightarrow q \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$

$p \rightarrow q \Leftrightarrow \neg p \vee q$

Show that $p \leftrightarrow q \Leftrightarrow (p \wedge q) \vee (\neg p \wedge \neg q)$ using logic algebra

$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws

$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

Example 4

$p \leftrightarrow q \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$
$p \rightarrow q \Leftrightarrow \neg p \vee q$

Show that $p \leftrightarrow q \Leftrightarrow (p \wedge q) \vee (\neg p \wedge \neg q)$ using logic algebra

$p \leftrightarrow q = (p \rightarrow q) \wedge (q \rightarrow p)$

$= (\neg p \vee q) \wedge (\neg q \vee p) = x \wedge (\neg q \vee p)$ $\quad (x = (\neg p \vee q))$

$= (x \wedge \neg q) \vee (x \wedge p)$ $\quad$ (distribution)

$= [(\neg p \vee q) \wedge \neg q] \vee (\neg p \vee q) \wedge p)]$

$= [(\neg p \wedge \neg q) \vee (q \wedge \neg q)] \vee [(\neg p \wedge p) \vee (q \wedge p)]$ $\quad$ (distribution)

$= [(\neg p \wedge \neg q) \vee F] \vee [F \vee (q \wedge p)]$

$= (\neg p \wedge \neg q) \vee (q \wedge p) = (p \wedge q) \vee (\neg p \wedge \neg q)$ $\quad$ (commutative)

Review Questions

Problem 1

$p \leftrightarrow q \Leftrightarrow (p \to q) \wedge (q \to p)$

$p \to q \Leftrightarrow \neg p \vee q$

- **Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent.**

$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws

$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

Problem 1

- Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent by developing a series of logical equivalences.

$$
\begin{aligned}
\neg(p\vee(\neg p\wedge q)) &\equiv \neg p \wedge \neg(\neg p\wedge q) && \text{By 2nd DeMorgan's} \\
&\equiv \neg p \wedge(\neg(\neg p)\vee\neg q) && \text{By 1st DeMorgan's} \\
&\equiv \neg p\wedge(p\vee\neg q) && \text{By double negation} \\
&\equiv (\neg p\wedge p) \vee (\neg p\wedge\neg q) && \text{By 2nd distributive} \\
&\equiv F \vee (\neg p\wedge\neg q) && \text{By definition of } \wedge \\
&\equiv (\neg p\wedge\neg q) \vee F && \text{By commutative law} \\
&\equiv (\neg p\wedge\neg q) && \text{By identity law for F}
\end{aligned}
$$

Problem 2

- Show p v (q ^ r) ≡ (p v q) ^ (p v r) using Truth Table

p	q	r	$q \wedge r$	$p \vee q$	$p \vee r$	$p \vee (q \wedge r)$	$(p \vee q) \wedge (p \vee r)$
F	F	F	F	F	F	F	F
F	F	T	F	F	T	F	F
F	T	F	F	T	F	F	F
F	T	T	T	T	T	T	T
T	F	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	T	F	F	T	T	T	T
T	T	T	T	T	T	T	T

Problem 3: Biconditional operator

Show $p \leftrightarrow q \Leftrightarrow (p \to q) \land (q \to p)$ using Truth Table

p	q	$p \leftrightarrow q$	$(p \to q)$	$(q \to p)$	$(p \to q) \land (q \to p)$
F	F	T	T	T	T
F	T	F	T	F	F
T	F	F	F	T	F
T	T	T	T	T	T

Try $p \leftrightarrow q \Leftrightarrow \neg(p \oplus q)$

Summary of Lecture: Propositional Logic

- Atomic propositions: p, q, r, …
- Boolean operators: $\neg \wedge \vee \oplus \rightarrow \leftrightarrow$
- Compound propositions: $s :\equiv (p \wedge \neg q) \vee r$
- Equivalences: $p \wedge \neg q \Leftrightarrow \neg(p \rightarrow q)$
- Proving equivalences using:
  - Truth tables
  - Symbolic derivations. $p \Leftrightarrow q \Leftrightarrow r$ …

Equivalence Rules

Equivalence	Name
$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws

Equivalence	Name
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

Precedence of Logical Operators

Formal Name	Nickname	Arity	Symbol
Negation operator	NOT	Unary	$\neg$
Conjunction operator	AND	Binary	$\wedge$
Disjunction operator	OR	Binary	$\vee$
Exclusive-OR operator	XOR	Binary	$\oplus$
Implication operator	IMPLIES	Binary	$\rightarrow$
Biconditional operator	IFF	Binary	$\leftrightarrow$

Supplementary Material

- Kimberly Brehm Channel
  - Discrete Math 1.2.1 - Translating Propositional Logic Statements
  - Discrete Math - 1.1.2 Implications Converse, Inverse, Contrapositive and Biconditionals

See you next lecture